DA-Practice-5.R

r1701177

2026-06-20

```
install.packages("vegan")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.6'
## (as 'lib' is unspecified)
```

```
library(vegan)
```

```
## Loading required package: permute
```

```
#--------------------start-------------------------------
# Get current working directory
getwd()
```

```
## [1] "/cloud/project"
```

```
#----------------read dataset----------------------------
data<-read.table("data.txt",header=TRUE,sep="\t",check.names = FALSE)
summary (data)
```

```
##         lake    Chrysosphaerella brevispina C. coronacircumspina C. longispina
##  Length    : 6   Min.   :0.0000             Min.   :0.0000       Min.   :0.0
##  N.unique : 6   1st Qu.:0.0000             1st Qu.:0.0000       1st Qu.:0.0
##  N.blank  : 0   Median :0.0000             Median :0.0000       Median :0.0
##  Min.nchar:11   Mean   :0.3333             Mean   :0.6667       Mean   :0.5
##  Max.nchar:31   3rd Qu.:0.7500             3rd Qu.:0.7500       3rd Qu.:0.0
##                 Max.   :1.0000             Max.   :3.0000       Max.   :3.0
##  Chrysosphaerella sp. Paraphysomonas bandaiensis  P. gladiata    P. vestita
##  Min.   :0.0000       Min.   :0.0000              Min.   :0.0   Min.   :0.0
##  1st Qu.:0.0000       1st Qu.:0.0000              1st Qu.:0.0   1st Qu.:0.0
##  Median :0.0000       Median :0.0000              Median :0.5   Median :0.5
##  Mean   :0.1667       Mean   :0.1667              Mean   :0.5   Mean   :0.5
##  3rd Qu.:0.0000       3rd Qu.:0.0000              3rd Qu.:1.0   3rd Qu.:1.0
##  Max.   :1.0000       Max.   :1.0000              Max.   :1.0   Max.   :1.0
##  Paraphysomonas sp. Mallomonas akrokomos      \xcc. alata f. alata
##  Min.   :0.0000     Min.   :0.000        Min.   :0.0000
##  1st Qu.:0.0000     1st Qu.:0.250        1st Qu.:0.0000
##  Median :0.0000     Median :1.000        Median :0.5000
##  Mean   :0.1667     Mean   :1.167        Mean   :0.6667
##  3rd Qu.:0.0000     3rd Qu.:1.750        3rd Qu.:1.0000
##  Max.   :1.0000     Max.   :3.000        Max.   :2.0000
##  M. alata f. hualvensis   M. caudata  M. crassisquama
##  Min.   :0.0000         Min.   :0.0   Min.   :0.000
##  1st Qu.:0.0000         1st Qu.:0.0   1st Qu.:1.000
##  Median :0.0000         Median :0.5   Median :1.000
##  Mean   :0.1667         Mean   :0.5   Mean   :1.167
```

```
##  3rd Qu.:0.0000         3rd Qu.:1.0   3rd Qu.:1.750
##  Max.   :1.0000         Max.   :1.0   Max.   :2.000
##  M. crassiquama var. papillosa M. heterospina    M. insignis
##  Min.   :0.00                  Min.   :0.0000   Min.   :0.0000
##  1st Qu.:0.00                  1st Qu.:0.0000   1st Qu.:0.0000
##  Median :0.50                  Median :0.0000   Median :0.5000
##  Mean   :1.00                  Mean   :0.6667   Mean   :0.6667
##  3rd Qu.:1.75                  3rd Qu.:1.5000   3rd Qu.:1.0000
##  Max.   :3.00                  Max.   :2.0000   Max.   :2.0000
##  M. striata var. striata  M. tonsurata    Chrysodidymus synuroideus
##  Min.   :0.0             Min.   :0.0000   Min.   :0.0000
##  1st Qu.:0.0             1st Qu.:0.0000   1st Qu.:0.0000
##  Median :0.5             Median :0.5000   Median :0.0000
##  Mean   :0.5             Mean   :0.8333   Mean   :0.1667
##  3rd Qu.:1.0             3rd Qu.:1.0000   3rd Qu.:0.0000
##  Max.   :1.0             Max.   :3.0000   Max.   :1.0000
##  Spiniferomonas abei S. bilacunosa    S. bourrellyi      S. cornuta
##  Min.   :0.0000      Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
##  1st Qu.:0.0000      1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
##  Median :0.0000      Median :0.5000   Median :0.5000   Median :0.5000
##  Mean   :0.3333      Mean   :0.8333   Mean   :0.8333   Mean   :0.8333
##  3rd Qu.:0.7500      3rd Qu.:1.7500   3rd Qu.:1.7500   3rd Qu.:1.0000
##  Max.   :1.0000      Max.   :2.0000   Max.   :2.0000   Max.   :3.0000
##   S. crusigera      S. serrata     S. triangularis  S. trioralis f. trioralis
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :1.000
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:1.000
##  Median :0.0000   Median :0.0000   Median :0.0000   Median :1.000
##  Mean   :0.3333   Mean   :0.3333   Mean   :0.1667   Mean   :1.333
##  3rd Qu.:0.0000   3rd Qu.:0.7500   3rd Qu.:0.0000   3rd Qu.:1.750
##  Max.   :2.0000   Max.   :1.0000   Max.   :1.0000   Max.   :2.000
##  S. trioralis f. cuspidata Spiniferomonas  sp.
##  Min.   :0.0000            Min.   :0.0000
##  1st Qu.:0.0000            1st Qu.:0.0000
##  Median :0.0000            Median :0.0000
##  Mean   :0.6667            Mean   :0.1667
##  3rd Qu.:1.5000            3rd Qu.:0.0000
##  Max.   :2.0000            Max.   :1.0000
##  Synura echinulata f. leptorrhabda S. petersenii f. petersenii
##  Min.   :0.000                     Min.   :0.0000
##  1st Qu.:0.250                     1st Qu.:0.0000
##  Median :1.500                     Median :0.0000
##  Mean   :1.167                     Mean   :0.6667
##  3rd Qu.:2.000                     3rd Qu.:1.5000
##  Max.   :2.000                     Max.   :2.0000
##  S. petersenii f. kufferathii S. punctulosa      S. spinosa
##  Min.   :0.00                 Min.   :0.0000   Min.   :0.0000
##  1st Qu.:0.00                 1st Qu.:0.0000   1st Qu.:0.2500
##  Median :0.50                 Median :0.0000   Median :1.0000
##  Mean   :1.00                 Mean   :0.1667   Mean   :0.8333
##  3rd Qu.:1.75                 3rd Qu.:0.0000   3rd Qu.:1.0000
##  Max.   :3.00                 Max.   :1.0000   Max.   :2.0000
```

```
data
```

```
##                                 lake Chrysosphaerella brevispina
```

```
## 1                  Ledyanaya Gora                           0
## 2                  Karaul village                           0
## 3               Ladyginskie Yary                            1
## 4              Sopochnaya Karga                             1
## 5 Sibiryakov Island, Srednee Lake                           0
## 6                     Chernyi Bay                           0
##   C. coronacircumspina C. longispina Chrysosphaerella sp.
## 1                    1             0                    0
## 2                    0             0                    0
## 3                    0             0                    0
## 4                    3             3                    1
## 5                    0             0                    0
## 6                    0             0                    0
##   Paraphysomonas bandaiensis P. gladiata P. vestita Paraphysomonas sp.
## 1                          0           1          0                  0
## 2                          0           0          0                  0
## 3                          1           0          1                  1
## 4                          0           1          1                  0
## 5                          0           0          0                  0
## 6                          0           1          1                  0
##   Mallomonas akrokomos \xcc. alata f. alata M. alata f. hualvensis M. caudata
## 1                    1                    1                      0          0
## 2                    0                    1                      0          1
## 3                    1                    0                      0          1
## 4                    3                    2                      0          0
## 5                    0                    0                      1          1
## 6                    2                    0                      0          0
##   M. crassisquama M. crassiquama var. papillosa M. heterospina M. insignis
## 1               1                             2              0           0
## 2               1                             0              0           0
## 3               1                             1              0           2
## 4               2                             0              0           1
## 5               0                             0              2           1
## 6               2                             3              2           0
##   M. striata var. striata M. tonsurata Chrysodidymus synuroideus
## 1                       0            3                         0
## 2                       0            1                         0
## 3                       1            0                         0
## 4                       1            0                         0
## 5                       1            1                         0
## 6                       0            0                         1
##   Spiniferomonas abei S. bilacunosa S. bourrellyi S. cornuta S. crusigera
## 1                   1             2             2          3            2
## 2                   0             0             0          0            0
## 3                   0             0             0          0            0
## 4                   1             2             2          1            0
## 5                   0             0             0          0            0
## 6                   0             1             1          1            0
##   S. serrata S. triangularis S. trioralis f. trioralis
## 1          0               1                         2
## 2          0               0                         1
## 3          1               0                         1
## 4          1               0                         2
## 5          0               0                         1
```

```
## 6          0               0                         1
##   S. trioralis f. cuspidata Spiniferomonas  sp.
## 1                         0                   0
## 2                         0                   0
## 3                         2                   0
## 4                         2                   1
## 5                         0                   0
## 6                         0                   0
##   Synura echinulata f. leptorrhabda S. petersenii f. petersenii
## 1                                 0                           0
## 2                                 0                           0
## 3                                 2                           2
## 4                                 2                           0
## 5                                 1                           0
## 6                                 2                           2
##   S. petersenii f. kufferathii S. punctulosa S. spinosa
## 1                            0             0          0
## 2                            0             0          1
## 3                            2             0          0
## 4                            0             1          1
## 5                            1             0          1
## 6                            3             0          2
```

```
rownames(data)<-data[,1]
data<-data[,-1]
data
```

```
##                                   Chrysosphaerella brevispina
## Ledyanaya Gora                                              0
## Karaul village                                              0
## Ladyginskie Yary                                            1
## Sopochnaya Karga                                            1
## Sibiryakov Island, Srednee Lake                             0
## Chernyi Bay                                                 0
##                                   C. coronacircumspina C. longispina
## Ledyanaya Gora                                       1             0
## Karaul village                                       0             0
## Ladyginskie Yary                                     0             0
## Sopochnaya Karga                                     3             3
## Sibiryakov Island, Srednee Lake                      0             0
## Chernyi Bay                                          0             0
##                                   Chrysosphaerella sp. Paraphysomonas bandaiensis
## Ledyanaya Gora                                       0                          0
## Karaul village                                       0                          0
## Ladyginskie Yary                                     0                          1
## Sopochnaya Karga                                     1                          0
## Sibiryakov Island, Srednee Lake                      0                          0
## Chernyi Bay                                          0                          0
##                                   P. gladiata P. vestita Paraphysomonas sp.
## Ledyanaya Gora                              1          0                  0
## Karaul village                              0          0                  0
## Ladyginskie Yary                            0          1                  1
## Sopochnaya Karga                            1          1                  0
## Sibiryakov Island, Srednee Lake             0          0                  0
## Chernyi Bay                                 1          1                  0
```

```
##                                  Mallomonas akrokomos \xcc. alata f. alata
## Ledyanaya Gora                                      1                   1
## Karaul village                                      0                   1
## Ladyginskie Yary                                    1                   0
## Sopochnaya Karga                                    3                   2
## Sibiryakov Island, Srednee Lake                     0                   0
## Chernyi Bay                                         2                   0
##                                  M. alata f. hualvensis M. caudata
## Ledyanaya Gora                                        0          0
## Karaul village                                        0          1
## Ladyginskie Yary                                      0          1
## Sopochnaya Karga                                      0          0
## Sibiryakov Island, Srednee Lake                       1          1
## Chernyi Bay                                           0          0
##                                  M. crassisquama M. crassiquama var. papillosa
## Ledyanaya Gora                                 1                             2
## Karaul village                                 1                             0
## Ladyginskie Yary                               1                             1
## Sopochnaya Karga                               2                             0
## Sibiryakov Island, Srednee Lake                0                             0
## Chernyi Bay                                    2                             3
##                                  M. heterospina M. insignis
## Ledyanaya Gora                                0           0
## Karaul village                                0           0
## Ladyginskie Yary                              0           2
## Sopochnaya Karga                              0           1
## Sibiryakov Island, Srednee Lake               2           1
## Chernyi Bay                                   2           0
##                                  M. striata var. striata M. tonsurata
## Ledyanaya Gora                                         0            3
## Karaul village                                         0            1
## Ladyginskie Yary                                       1            0
## Sopochnaya Karga                                       1            0
## Sibiryakov Island, Srednee Lake                        1            1
## Chernyi Bay                                            0            0
##                                  Chrysodidymus synuroideus Spiniferomonas abei
## Ledyanaya Gora                                           0                   1
## Karaul village                                           0                   0
## Ladyginskie Yary                                         0                   0
## Sopochnaya Karga                                         0                   1
## Sibiryakov Island, Srednee Lake                          0                   0
## Chernyi Bay                                              1                   0
##                                  S. bilacunosa S. bourrellyi S. cornuta
## Ledyanaya Gora                               2             2          3
## Karaul village                               0             0          0
## Ladyginskie Yary                             0             0          0
## Sopochnaya Karga                             2             2          1
## Sibiryakov Island, Srednee Lake              0             0          0
## Chernyi Bay                                  1             1          1
##                                  S. crusigera S. serrata S. triangularis
## Ledyanaya Gora                              2          0               1
## Karaul village                              0          0               0
## Ladyginskie Yary                            0          1               0
## Sopochnaya Karga                            0          1               0
```

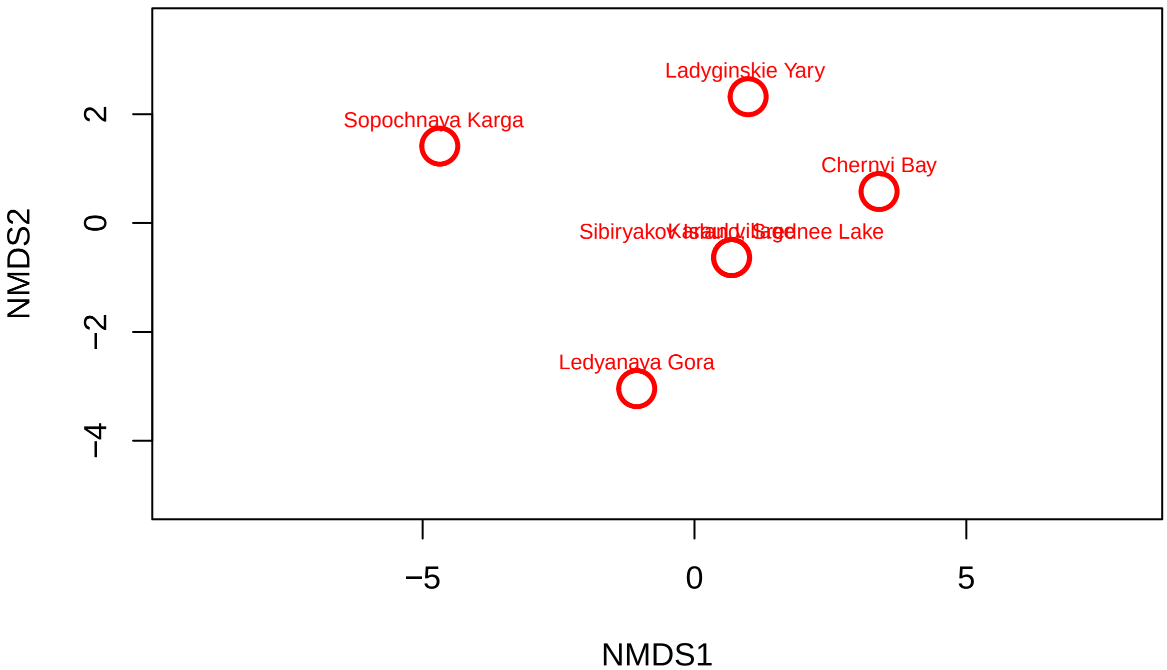
```
## Sibiryakov Island, Srednee Lake           0          0               0
## Chernyi Bay                               0          0               0
##                                 S. trioralis f. trioralis
## Ledyanaya Gora                                          2
## Karaul village                                          1
## Ladyginskie Yary                                        1
## Sopochnaya Karga                                        2
## Sibiryakov Island, Srednee Lake                         1
## Chernyi Bay                                             1
##                                 S. trioralis f. cuspidata Spiniferomonas  sp.
## Ledyanaya Gora                                          0                   0
## Karaul village                                          0                   0
## Ladyginskie Yary                                        2                   0
## Sopochnaya Karga                                        2                   1
## Sibiryakov Island, Srednee Lake                         0                   0
## Chernyi Bay                                             0                   0
##                                 Synura echinulata f. leptorrhabda
## Ledyanaya Gora                                                  0
## Karaul village                                                  0
## Ladyginskie Yary                                                2
## Sopochnaya Karga                                                2
## Sibiryakov Island, Srednee Lake                                 1
## Chernyi Bay                                                     2
##                                 S. petersenii f. petersenii
## Ledyanaya Gora                                            0
## Karaul village                                            0
## Ladyginskie Yary                                          2
## Sopochnaya Karga                                          0
## Sibiryakov Island, Srednee Lake                           0
## Chernyi Bay                                               2
##                                 S. petersenii f. kufferathii S. punctulosa
## Ledyanaya Gora                                             0             0
## Karaul village                                             0             0
## Ladyginskie Yary                                           2             0
## Sopochnaya Karga                                           0             1
## Sibiryakov Island, Srednee Lake                            1             0
## Chernyi Bay                                                3             0
##                                 S. spinosa
## Ledyanaya Gora                           0
## Karaul village                           1
## Ladyginskie Yary                         0
## Sopochnaya Karga                         1
## Sibiryakov Island, Srednee Lake          1
## Chernyi Bay                              2
```

```r
# Non-metric multidimensional scaling (NMDS) using Euclidean distances
# Perform NMDS ordination based on Euclidean distances.
# Euclidean distance is suitable for quantitative data but may be less appropriate
# for ecological community data due to sensitivity to double zeros.
# The metaMDS function automatically runs multiple random starts and scales the solution.
ord <- metaMDS(data, distance = "euclidean")
```

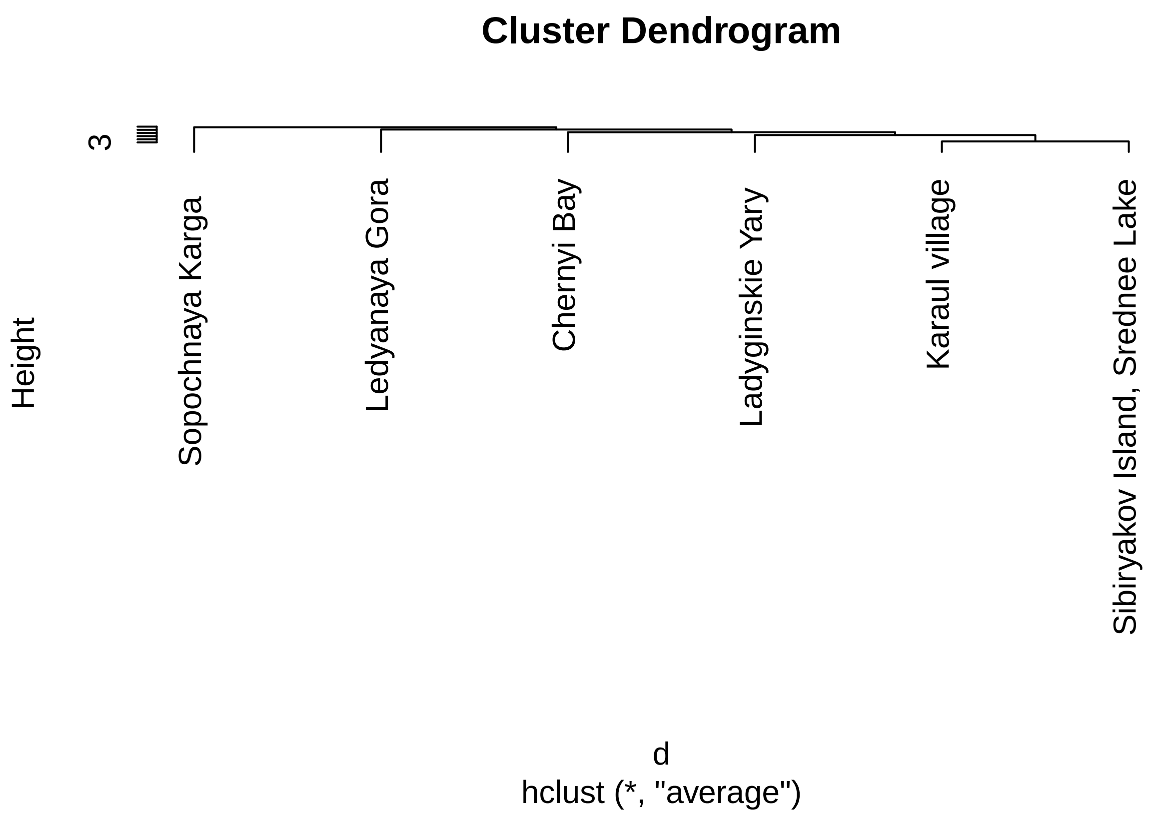
```
## Run 0 stress 0.0848674
## Run 1 stress 0.1620205
## Run 2 stress 0.1201528
```

```
## Run 3 stress 9.734365e-05
## ... New best solution
## ... Procrustes: rmse 0.1910603  max resid 0.3203572
## Run 4 stress 0.04685852
## Run 5 stress 8.990827e-05
## ... New best solution
## ... Procrustes: rmse 1.78297e-05  max resid 2.447792e-05
## ... Similar to previous best
## Run 6 stress 0.276134
## Run 7 stress 9.652409e-05
## ... Procrustes: rmse 0.0002663948  max resid 0.0005847019
## ... Similar to previous best
## Run 8 stress 0.04685852
## Run 9 stress 9.241977e-05
## ... Procrustes: rmse 0.2263286  max resid 0.2898795
## Run 10 stress 8.636866e-05
## ... New best solution
## ... Procrustes: rmse 0.2263337  max resid 0.2898953
## Run 11 stress 9.593838e-05
## ... Procrustes: rmse 0.2261889  max resid 0.3451839
## Run 12 stress 0.04685852
## Run 13 stress 0.04685852
## Run 14 stress 8.009426e-05
## ... New best solution
## ... Procrustes: rmse 0.2263247  max resid 0.3454036
## Run 15 stress 0.1620205
## Run 16 stress 9.774572e-05
## ... Procrustes: rmse 0.0002879694  max resid 0.000632579
## ... Similar to previous best
## Run 17 stress 0.04685852
## Run 18 stress 0.04685852
## Run 19 stress 0.1201528
## Run 20 stress 8.036563e-05
## ... Procrustes: rmse 0.0002564779  max resid 0.0005658402
## ... Similar to previous best
## *** Best solution repeated 2 times

## Warning in metaMDS(data, distance = "euclidean"): stress is (nearly) zero: you
## may have insufficient data
```

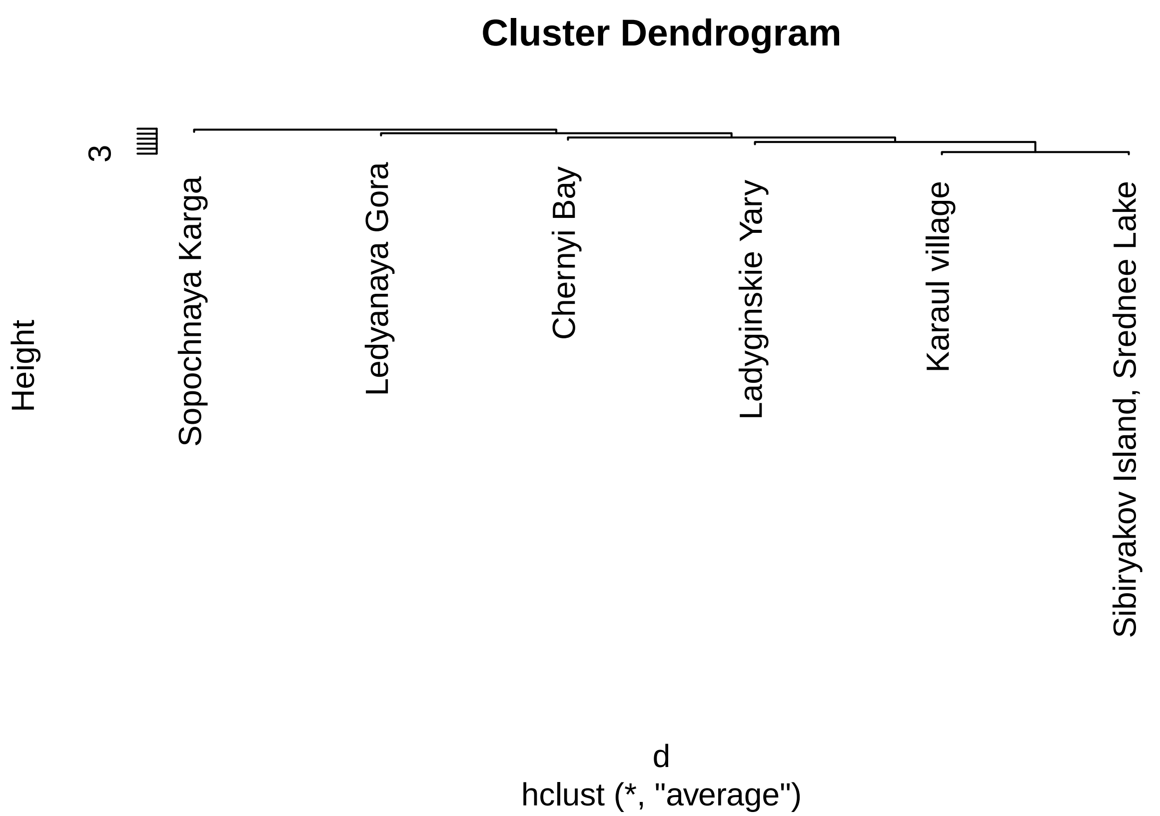
```r
#Visualisation of NMDS results
plot(ord, type = "n")                        # Create empty plot (axes only)
points(ord, disp = "sites", pch = 21,
       cex = 2.5, lwd = 2.5, col = "red") # Add site points (samples)
text(ord, display = "site", cex = 0.7,
     col = "red", pos = 3)                         # Label sites with their names
```




```r
# Cluster analysis with Euclidean distances
# Compute Euclidean distance matrix between sites
d<-vegdist(data,method="euclidean")
# Perform hierarchical agglomerative clustering using average linkage (UPGMA)
fit<-hclust(d, method="average")

# Visualise the dendrogram with labels aligned at the baseline
plot(fit, hang =-1)
```



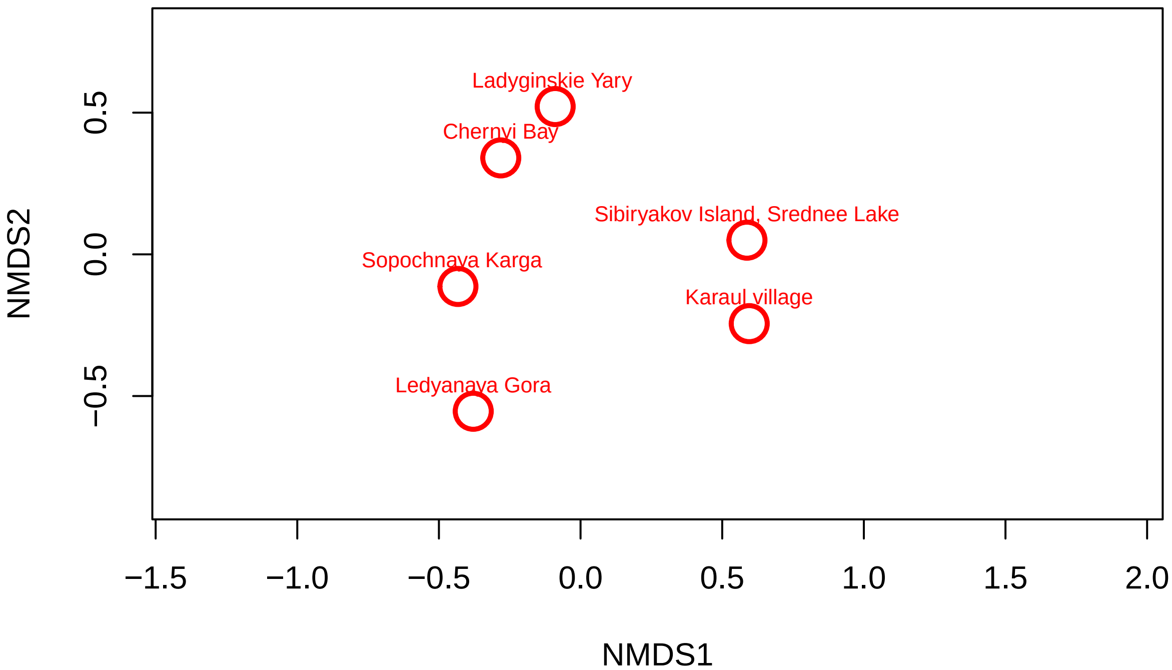

```r
# Simple dendrogram plot (default parameters)
plot(fit)
```



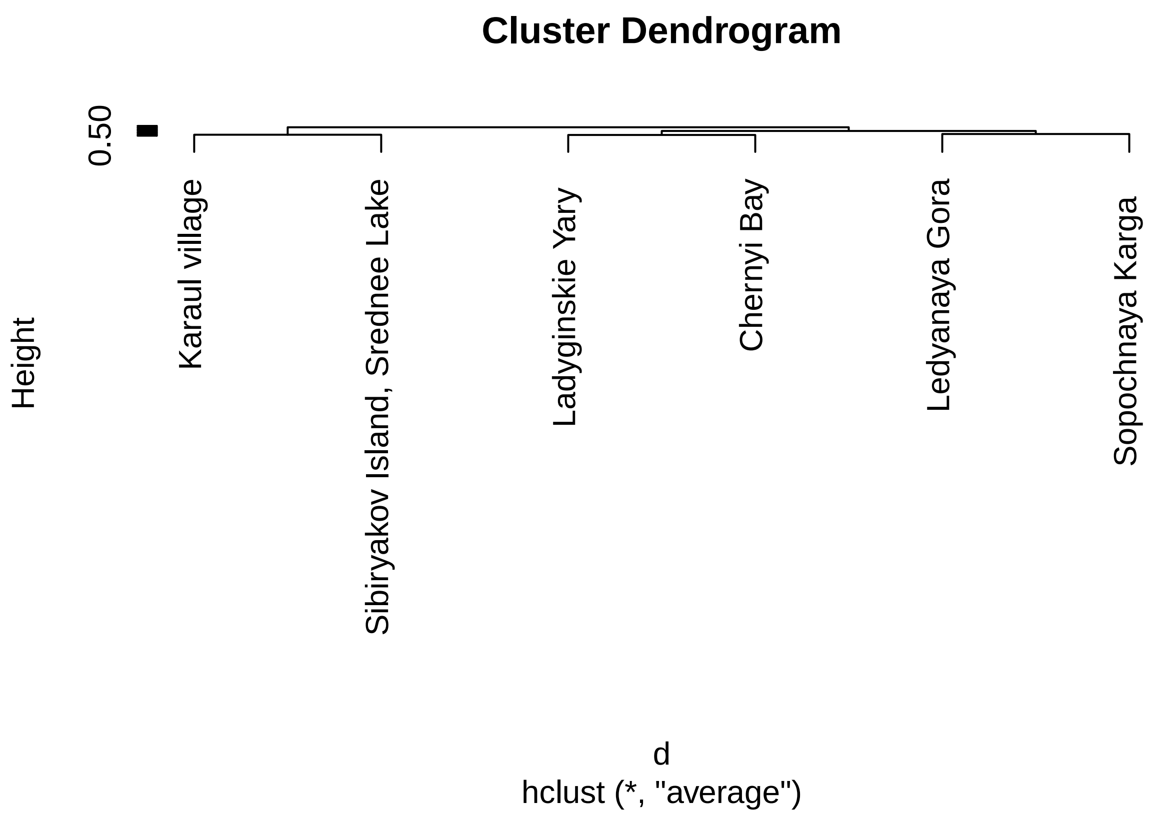
```r
#///////////////////////////////////////////////////////////////
#Non-metric multidimensional scaling (NMDS) with Bray-Curtis dissimilarities
# Bray-Curtis is a standard ecological distance measure that ignores double zeros
# and is robust for abundance data.
# The metaMDS function automatically runs multiple random starts and scales the final solution.
ord <- metaMDS(data, distance = "bray")
```

```
## Run 0 stress 0
## Run 1 stress 0.1499987
## Run 2 stress 0.1680323
## Run 3 stress 0
## ... Procrustes: rmse 0.02895574  max resid 0.05319101
## Run 4 stress 0.1815497
## Run 5 stress 4.426001e-05
## ... Procrustes: rmse 0.026573  max resid 0.03658767
## Run 6 stress 9.273874e-05
## ... Procrustes: rmse 0.06339456  max resid 0.1075813
## Run 7 stress 0
## ... Procrustes: rmse 0.01801475  max resid 0.02535063
## Run 8 stress 8.729379e-05
## ... Procrustes: rmse 0.08836139  max resid 0.1335126
## Run 9 stress 0.2076063
## Run 10 stress 0.2181459
## Run 11 stress 0.1815497
## Run 12 stress 3.072681e-05
## ... Procrustes: rmse 0.05066635  max resid 0.07776193
## Run 13 stress 0
## ... Procrustes: rmse 0.01619988  max resid 0.02391988
## Run 14 stress 0
## ... Procrustes: rmse 0.0255282  max resid 0.03602531
## Run 15 stress 1.51747e-05
## ... Procrustes: rmse 0.08024556  max resid 0.120545
## Run 16 stress 5.345661e-05
## ... Procrustes: rmse 0.04192804  max resid 0.06578049
## Run 17 stress 8.944003e-05
## ... Procrustes: rmse 0.02602655  max resid 0.0442601
## Run 18 stress 0
## ... Procrustes: rmse 0.06000663  max resid 0.1048761
## Run 19 stress 0.2076063
## Run 20 stress 5.336732e-05
## ... Procrustes: rmse 0.03694905  max resid 0.06385776
## *** Best solution was not repeated -- monoMDS stopping criteria:
##     13: stress < smin
##      4: stress ratio > sratmax
##      3: scale factor of the gradient < sfgrmin

## Warning in metaMDS(data, distance = "bray"): stress is (nearly) zero: you may
## have insufficient data
```

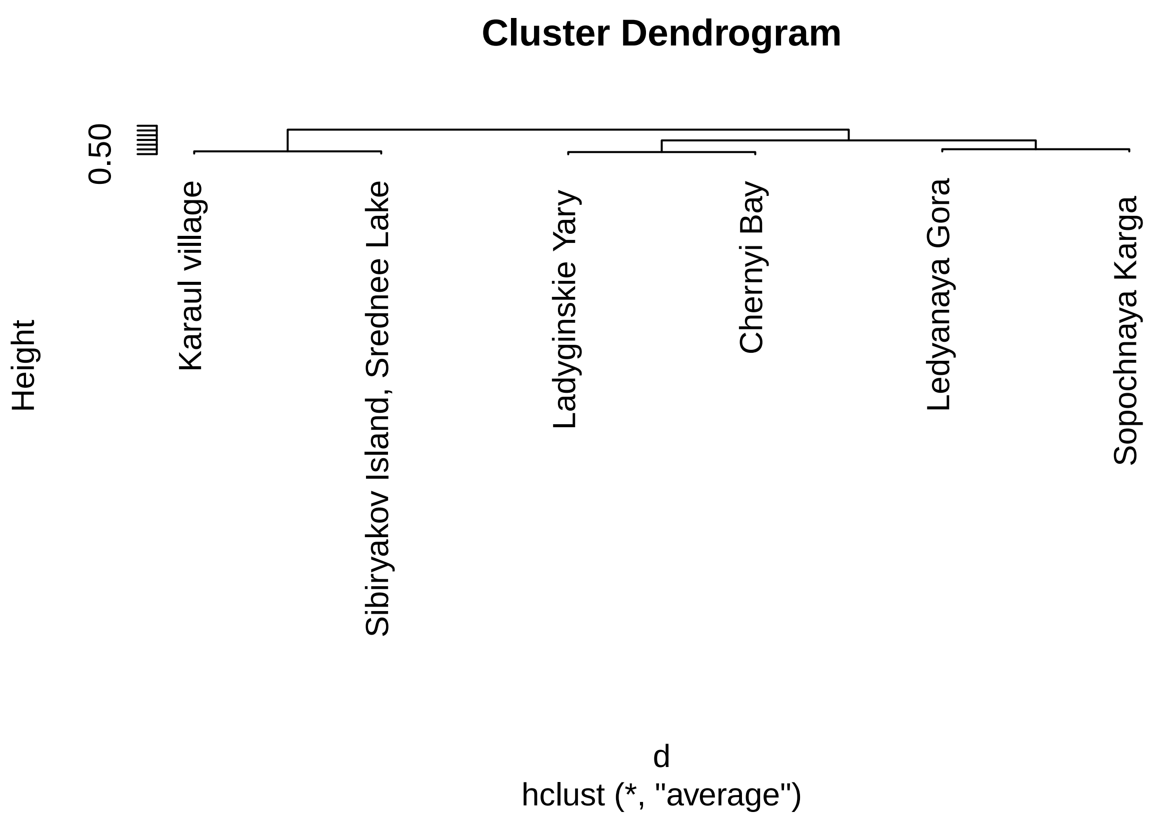
```r
#Visualisation of NMDS results
plot(ord, type = "n")
points(ord, disp="sites", pch=21, cex=2.5, lwd=2.5, col = "red")
text(ord, display = "site", cex=0.7, col="red", pos = 3)
```




```r
# Cluster analysis with Bray-Curtis distances
# Compute Bray-Curtis dissimilarity matrix (standard for ecological community data)
d<-vegdist(data,method="bray")
# Perform hierarchical agglomerative clustering using average linkage (UPGMA)
fit<-hclust(d, method="average")

# Visualise dendrogram with labels aligned at the same horizontal level
plot(fit, hang =-1)
```




```r
# Simple dendrogram plot (default R style)
plot(fit)
```



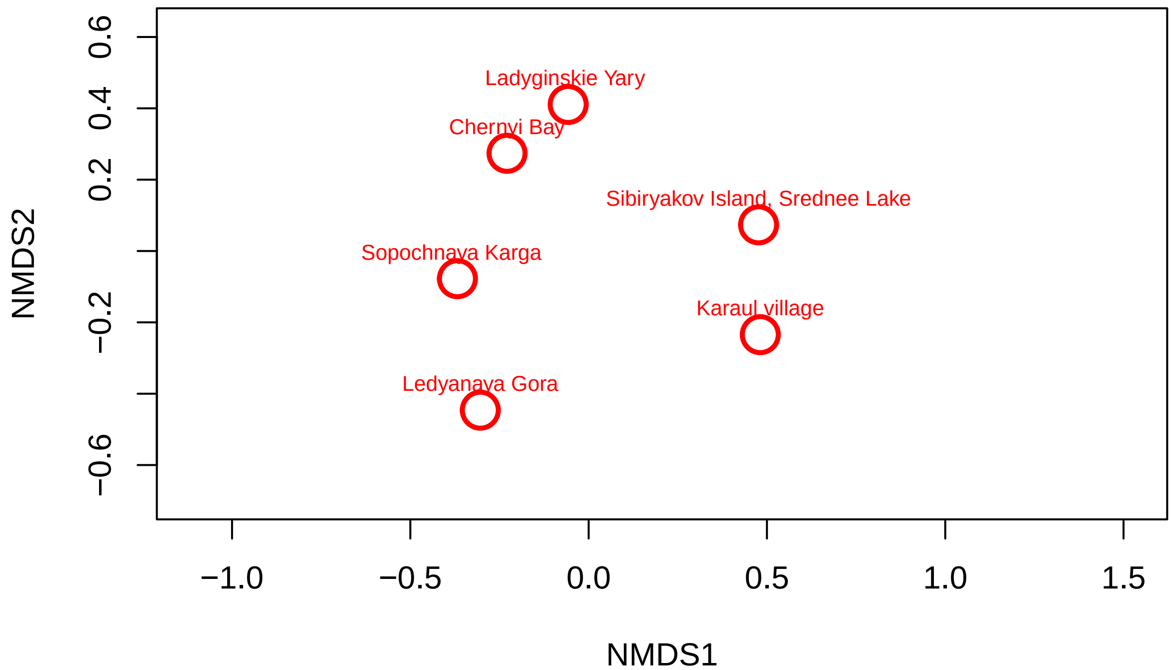
```r
#////////////////////////////////////////////////////////////////
# Non-metric multidimensional scaling (NMDS) with Jaccard distance
ord <- metaMDS(data, distance = "jaccard")
```

```
## Run 0 stress 0
## Run 1 stress 0
## ... Procrustes: rmse 0.02556189  max resid 0.04262108
## Run 2 stress 0.2181459
## Run 3 stress 0.1499987
## Run 4 stress 0
## ... Procrustes: rmse 0.01668198  max resid 0.0268855
## Run 5 stress 0.1680323
## Run 6 stress 0.1135296
## Run 7 stress 0.2269841
## Run 8 stress 8.935703e-05
## ... Procrustes: rmse 0.02131321  max resid 0.03054633
## Run 9 stress 9.541265e-05
## ... Procrustes: rmse 0.1040823  max resid 0.1640719
## Run 10 stress 0.1499987
## Run 11 stress 8.917537e-05
## ... Procrustes: rmse 0.1063174  max resid 0.163175
## Run 12 stress 8.122501e-05
## ... Procrustes: rmse 0.02291762  max resid 0.03916129
## Run 13 stress 0
## ... Procrustes: rmse 0.03050143  max resid 0.0371955
## Run 14 stress 0.2171934
## Run 15 stress 3.753842e-05
## ... Procrustes: rmse 0.08871485  max resid 0.1443663
## Run 16 stress 9.307878e-05
## ... Procrustes: rmse 0.1030893  max resid 0.1648984
## Run 17 stress 7.996062e-05
## ... Procrustes: rmse 0.01943047  max resid 0.0366335
## Run 18 stress 5.339889e-05
## ... Procrustes: rmse 0.03167698  max resid 0.05156659
## Run 19 stress 9.88064e-05
## ... Procrustes: rmse 0.1033499  max resid 0.1659388
## Run 20 stress 0.2181459
## *** Best solution was not repeated -- monoMDS stopping criteria:
##     12: stress < smin
##      2: stress ratio > sratmax
##      6: scale factor of the gradient < sfgrmin
```

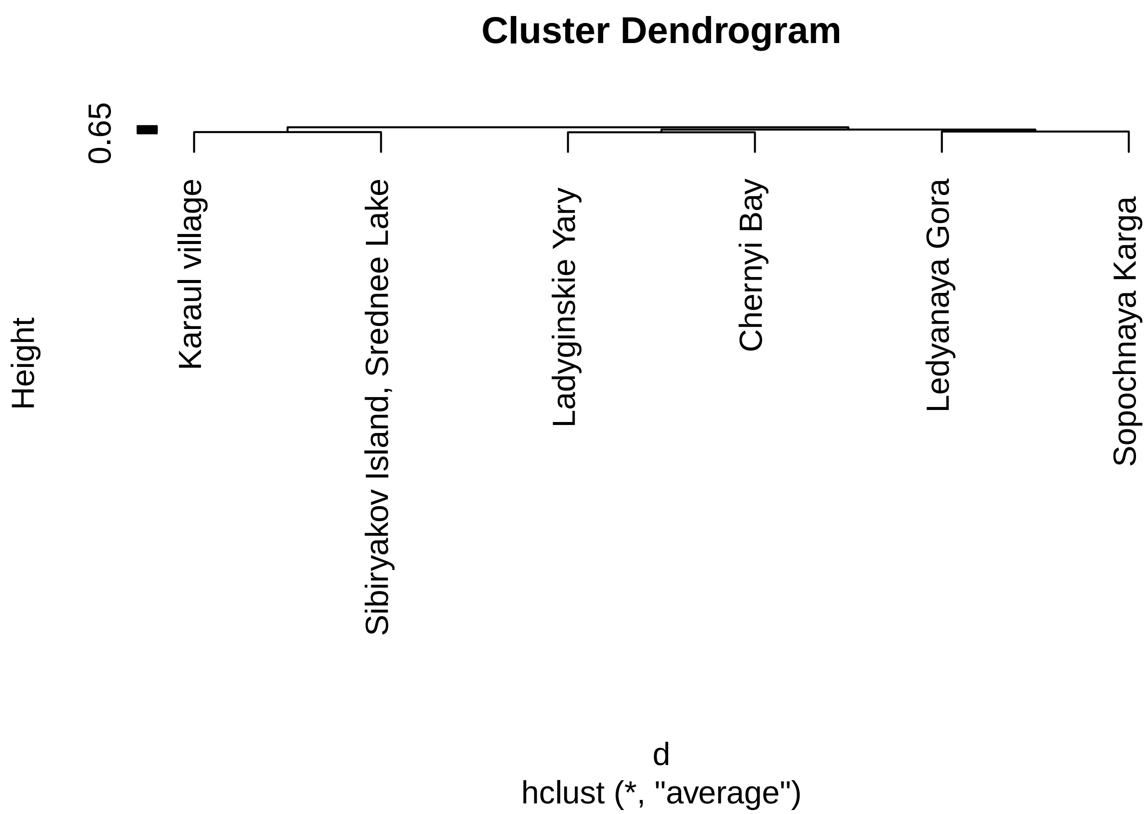
```
## Warning in metaMDS(data, distance = "jaccard"): stress is (nearly) zero: you
## may have insufficient data
```

```
## Warning in postMDS(out$points, dis, plot = max(0, plot - 1), ...): skipping
## half-change scaling: too few points below threshold
```

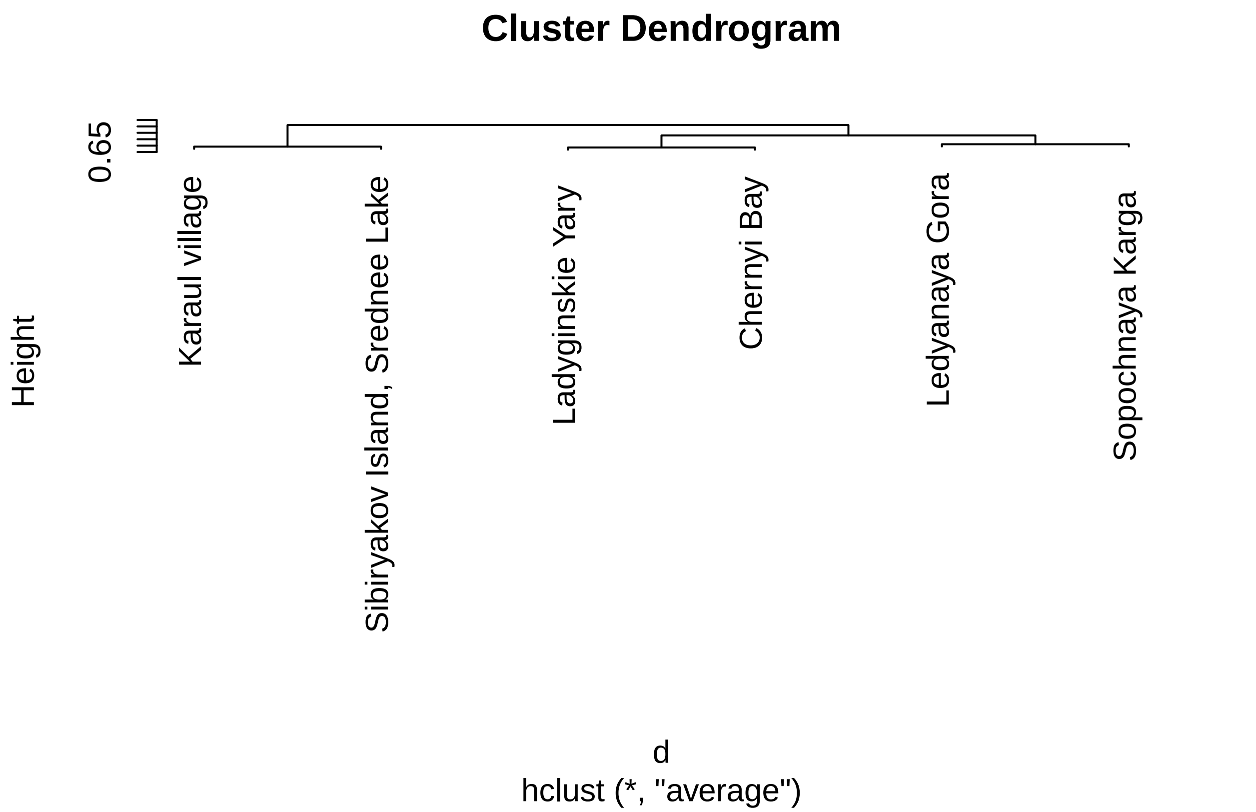
```r
plot(ord, type = "n")
points(ord, disp="sites", pch=21, cex=2.5, lwd=2.5, col = "red")
text(ord, display = "site", cex=0.7, col="red", pos = 3)
```




```r
# Cluster analysis with Jaccard distance
# Compute Jaccard dissimilarity matrix
d<-vegdist(data,method="jaccard")
# Hierarchical clustering using average linkage (UPGMA)
fit<-hclust(d, method="average")
# Visualise dendrogram with labels aligned
plot(fit, hang =-1)
```



```r
# Default dendrogram plot
plot(fit)
```




```r
#//////////////////////////////////////////////////////////////
# TASK 5: Detailed analysis with Bray-Curtis distance
#//////////////////////////////////////////////////////////////
```

```r
# STEP 1: Compute NMDS with Bray-Curtis dissimilarity
```

```r
set.seed(42)  # for reproducibility
ord <- metaMDS(data, distance = "bray", k = 2, trymax = 100)
```

```
## Run 0 stress 0
## Run 1 stress 0
## ... Procrustes: rmse 0.02308799  max resid 0.02974219
## Run 2 stress 0.2181459
## Run 3 stress 0.1815498
## Run 4 stress 0.1499987
## Run 5 stress 0.1815497
## Run 6 stress 9.357739e-05
## ... Procrustes: rmse 0.08811158  max resid 0.1325105
## Run 7 stress 0
## ... Procrustes: rmse 0.02925744  max resid 0.04458166
## Run 8 stress 0
## ... Procrustes: rmse 0.02151733  max resid 0.03051056
## Run 9 stress 0
## ... Procrustes: rmse 0.01878271  max resid 0.02483312
## Run 10 stress 0.2761338
## Run 11 stress 0
```

```
## ... Procrustes: rmse 0.03000018  max resid 0.04974647
## Run 12 stress 0.2181459
## Run 13 stress 8.994092e-05
## ... Procrustes: rmse 0.1198568  max resid 0.1859369
## Run 14 stress 0.2181459
## Run 15 stress 0.2076063
## Run 16 stress 0
## ... Procrustes: rmse 0.0218798  max resid 0.03370885
## Run 17 stress 0
## ... Procrustes: rmse 0.04783363  max resid 0.07072259
## Run 18 stress 0.1687717
## Run 19 stress 6.922235e-05
## ... Procrustes: rmse 0.03958808  max resid 0.06169729
## Run 20 stress 0
## ... Procrustes: rmse 0.01855941  max resid 0.02714016
## Run 21 stress 0
## ... Procrustes: rmse 0.0208832  max resid 0.03358733
## Run 22 stress 0.2181459
## Run 23 stress 0
## ... Procrustes: rmse 0.03670032  max resid 0.05289681
## Run 24 stress 0
## ... Procrustes: rmse 0.03314948  max resid 0.04705997
## Run 25 stress 7.16364e-05
## ... Procrustes: rmse 0.02728461  max resid 0.05098012
## Run 26 stress 9.607146e-05
## ... Procrustes: rmse 0.08959975  max resid 0.1311211
## Run 27 stress 9.756392e-05
## ... Procrustes: rmse 0.03847486  max resid 0.06773397
## Run 28 stress 0.2181459
## Run 29 stress 0.2761339
## Run 30 stress 0
## ... Procrustes: rmse 0.02917859  max resid 0.0443964
## Run 31 stress 0.1616427
## Run 32 stress 8.452042e-05
## ... Procrustes: rmse 0.0888361  max resid 0.1339118
## Run 33 stress 0.2076063
## Run 34 stress 0.1687717
## Run 35 stress 0.1815497
## Run 36 stress 0.1687717
## Run 37 stress 0.1499987
## Run 38 stress 0
## ... Procrustes: rmse 0.02915086  max resid 0.04006705
## Run 39 stress 9.346102e-05
## ... Procrustes: rmse 0.08843649  max resid 0.1336026
## Run 40 stress 0
## ... Procrustes: rmse 0.06292633  max resid 0.08220931
## Run 41 stress 0.1815498
## Run 42 stress 0.2076063
## Run 43 stress 8.167439e-05
## ... Procrustes: rmse 0.03843191  max resid 0.06521548
## Run 44 stress 0
## ... Procrustes: rmse 0.02990423  max resid 0.05437455
## Run 45 stress 0.2181459
## Run 46 stress 0.2171934
```

```
## Run 47 stress 0.2563318
## Run 48 stress 5.673182e-05
## ... Procrustes: rmse 0.1278577  max resid 0.1984258
## Run 49 stress 0.1680323
## Run 50 stress 0.1815497
## Run 51 stress 0
## ... Procrustes: rmse 0.02257451  max resid 0.04119995
## Run 52 stress 0.1135292
## Run 53 stress 0.1135292
## Run 54 stress 0.1687717
## Run 55 stress 0
## ... Procrustes: rmse 0.01519495  max resid 0.02215103
## Run 56 stress 0.1680323
## Run 57 stress 0
## ... Procrustes: rmse 0.01745115  max resid 0.02691134
## Run 58 stress 9.150809e-05
## ... Procrustes: rmse 0.08822887  max resid 0.1333418
## Run 59 stress 0.1135294
## Run 60 stress 0.1499987
## Run 61 stress 8.755482e-05
## ... Procrustes: rmse 0.0880581  max resid 0.132856
## Run 62 stress 0.1680323
## Run 63 stress 0
## ... Procrustes: rmse 0.01664295  max resid 0.02636311
## Run 64 stress 0
## ... Procrustes: rmse 0.01346592  max resid 0.01814799
## Run 65 stress 0.1499987
## Run 66 stress 4.047396e-05
## ... Procrustes: rmse 0.009283759  max resid 0.01296081
## Run 67 stress 9.918477e-05
## ... Procrustes: rmse 0.0905623  max resid 0.1306896
## Run 68 stress 6.210847e-05
## ... Procrustes: rmse 0.1330159  max resid 0.2050474
## Run 69 stress 0.1815497
## Run 70 stress 0
## ... Procrustes: rmse 0.04483873  max resid 0.05681139
## Run 71 stress 0
## ... Procrustes: rmse 0.05930585  max resid 0.08985647
## Run 72 stress 0.2181459
## Run 73 stress 9.232737e-05
## ... Procrustes: rmse 0.03531983  max resid 0.05570536
## Run 74 stress 9.963303e-05
## ... Procrustes: rmse 0.03978915  max resid 0.05632206
## Run 75 stress 0
## ... Procrustes: rmse 0.03356248  max resid 0.05387347
## Run 76 stress 0.1499987
## Run 77 stress 0.1680323
## Run 78 stress 0.1687717
## Run 79 stress 0.1499987
## Run 80 stress 0
## ... Procrustes: rmse 0.03534446  max resid 0.05704422
## Run 81 stress 0
## ... Procrustes: rmse 0.01849896  max resid 0.02633721
## Run 82 stress 0.1499987
```

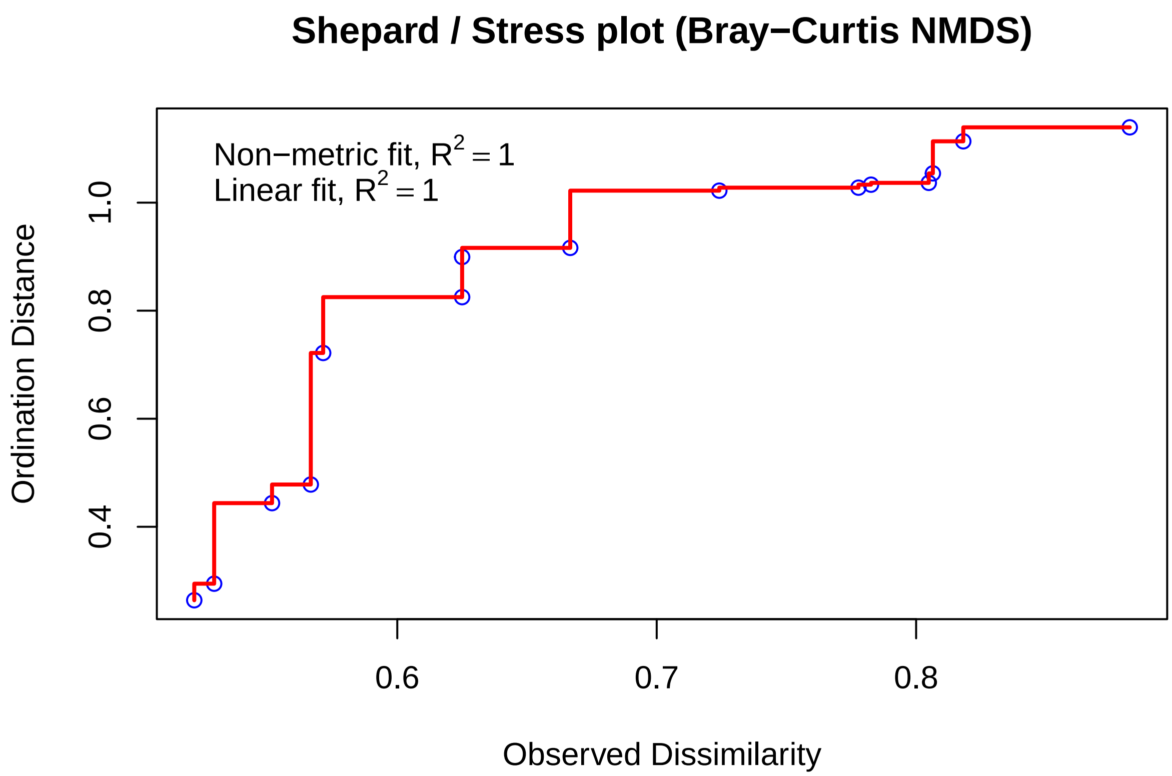
```
## Run 83 stress 7.154369e-05
## ... Procrustes: rmse 0.1350724  max resid 0.2077707
## Run 84 stress 0.2673072
## Run 85 stress 8.857115e-05
## ... Procrustes: rmse 0.08892198  max resid 0.1339792
## Run 86 stress 0
## ... Procrustes: rmse 0.03383659  max resid 0.05177866
## Run 87 stress 0.1616427
## Run 88 stress 0.1815497
## Run 89 stress 0.1687717
## Run 90 stress 0
## ... Procrustes: rmse 0.02737565  max resid 0.0410307
## Run 91 stress 0.1499987
## Run 92 stress 0
## ... Procrustes: rmse 0.02163809  max resid 0.0371202
## Run 93 stress 0
## ... Procrustes: rmse 0.01818235  max resid 0.02432908
## Run 94 stress 0
## ... Procrustes: rmse 0.01165615  max resid 0.01766096
## Run 95 stress 3.484539e-05
## ... Procrustes: rmse 0.09865282  max resid 0.1252207
## Run 96 stress 8.284032e-05
## ... Procrustes: rmse 0.04105565  max resid 0.07269862
## Run 97 stress 1.214913e-05
## ... Procrustes: rmse 0.05127603  max resid 0.08774882
## Run 98 stress 0.2269841
## Run 99 stress 9.317646e-05
## ... Procrustes: rmse 0.08828135  max resid 0.1334235
## Run 100 stress 0.2076063
## *** Best solution was not repeated -- monoMDS stopping criteria:
##     53: stress < smin
##     15: stress ratio > sratmax
##     32: scale factor of the gradient < sfgrmin
```

```
## Warning in metaMDS(data, distance = "bray", k = 2, trymax = 100): stress is
## (nearly) zero: you may have insufficient data
```

```r
# Print stress value
cat("NMDS stress:", ord$stress, "\n")
```

```
## NMDS stress: 0
```

```r
# Stressplot (Shepard diagram)
stressplot(ord, main = "Shepard / Stress plot (Bray-Curtis NMDS)")
```




```r
# STEP 2: Fit significant species vectors with envfit (p <= 0.05)
```

```r
fit_sp <- envfit(ord, data, permutations = 999, na.rm = TRUE)
```

```
## 'nperm' >= set of all permutations: complete enumeration.

## Set of permutations < 'minperm'. Generating entire set.

## 'nperm' >= set of all permutations: complete enumeration.

## Set of permutations < 'minperm'. Generating entire set.
```

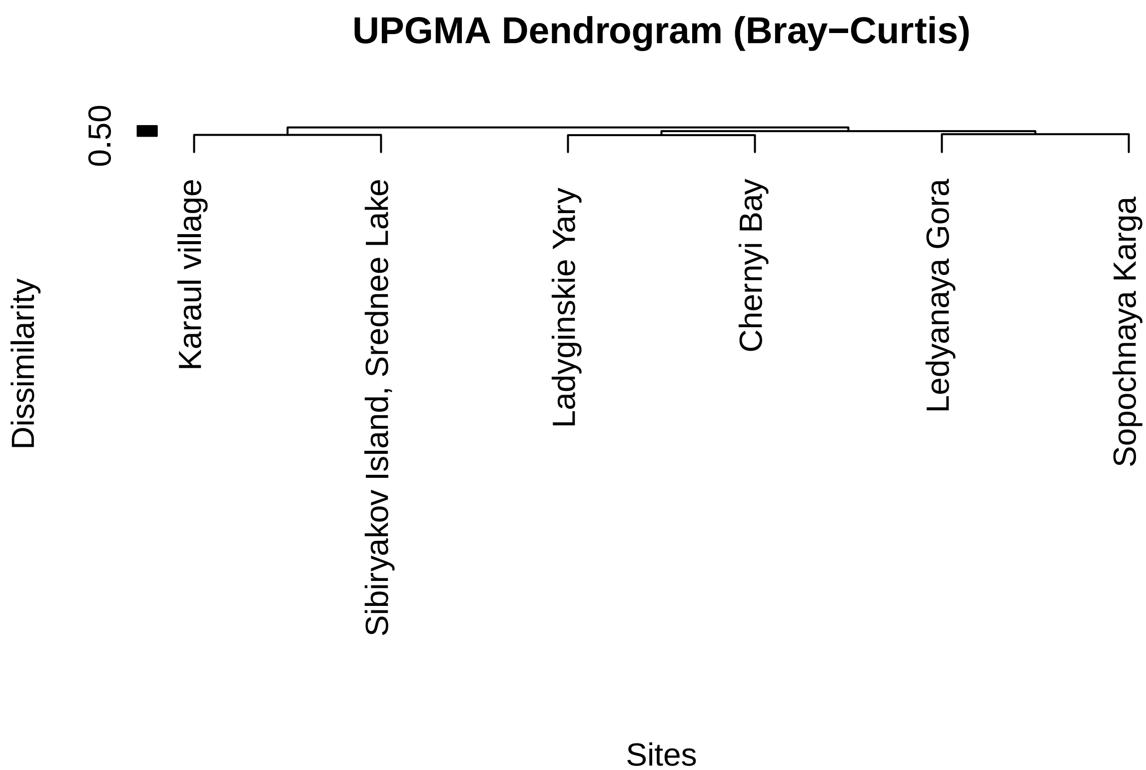
```r
print(fit_sp)
```

```
##
## ***VECTORS
##
##                                       NMDS1    NMDS2     r2  Pr(>r)
## Chrysosphaerella brevispina        -0.66121  0.75020 0.3455 0.46667
## C. coronacircumspina               -0.79077 -0.61212 0.4848 0.33333
## C. longispina                      -0.93464 -0.35558 0.2212 1.00000
## Chrysosphaerella sp.               -0.93464 -0.35558 0.2212 1.00000
## Paraphysomonas bandaiensis         -0.11751  0.99307 0.4327 0.50000
## P. gladiata                        -0.91724 -0.39833 0.8054 0.20000
## P. vestita                         -0.59468  0.80396 0.8708 0.10000 .
## Paraphysomonas sp.                 -0.11751  0.99307 0.4327 0.50000
## Mallomonas akrokomos               -0.98205  0.18864 0.7289 0.16667
## \xcc. alata f. alata               -0.39959 -0.91669 0.5227 0.36667
## M. alata f. hualvensis              0.99246  0.12254 0.3743 0.66667
## M. caudata                          0.91724  0.39833 0.8054 0.20000
## M. crassisquama                    -0.98106  0.19372 0.5495 0.31667
## M. crassiquama var. papillosa      -0.93688  0.34965 0.3509 0.52500
```

```
## M. heterospina                     0.47485  0.88007 0.2112 0.73333
## M. insignis                       -0.01733  0.99985 0.3742 0.50000
## M. striata var. striata            0.09720  0.99527 0.1843 0.70000
## M. tonsurata                       0.01722 -0.99985 0.6553 0.16667
## Chrysodidymus synuroideus         -0.49505  0.86887 0.2657 0.83333
## Spiniferomonas abei               -0.64211 -0.76661 0.8758 0.13333
## S. bilacunosa                     -0.79696 -0.60404 0.9373 0.06667 .
## S. bourrellyi                     -0.79696 -0.60404 0.9373 0.06667 .
## S. cornuta                        -0.66415 -0.74760 0.8398 0.03333 *
## S. crusigera                      -0.42607 -0.90469 0.6323 0.16667
## S. serrata                        -0.66121  0.75020 0.3455 0.46667
## S. triangularis                   -0.42607 -0.90469 0.6323 0.16667
## S. trioralis f. trioralis         -0.64211 -0.76661 0.8758 0.13333
## S. trioralis f. cuspidata         -0.66121  0.75020 0.3455 0.46667
## Spiniferomonas  sp.               -0.93464 -0.35558 0.2212 1.00000
## Synura echinulata f. leptorrhabda -0.41365  0.91044 0.8360 0.11667
## S. petersenii f. petersenii       -0.28434  0.95872 0.8162 0.20000
## S. petersenii f. kufferathii      -0.14081  0.99004 0.7471 0.15833
## S. punctulosa                     -0.93464 -0.35558 0.2212 1.00000
## S. spinosa                         0.32633  0.94526 0.0747 0.90000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## Permutation: free
## Number of permutations: 719
```

```r
# STEP 3: UPGMA hierarchical clustering with Bray-Curtis
```

```r
d_bray <- vegdist(data, method = "bray")
fit_clust <- hclust(d_bray, method = "average")  # UPGMA

plot(fit_clust, hang = -1,
     main = "UPGMA Dendrogram (Bray-Curtis)",
     xlab = "Sites", sub = "", ylab = "Dissimilarity")
```




```r
# STEP 4: Cut dendrogram into 2-3 clusters
```

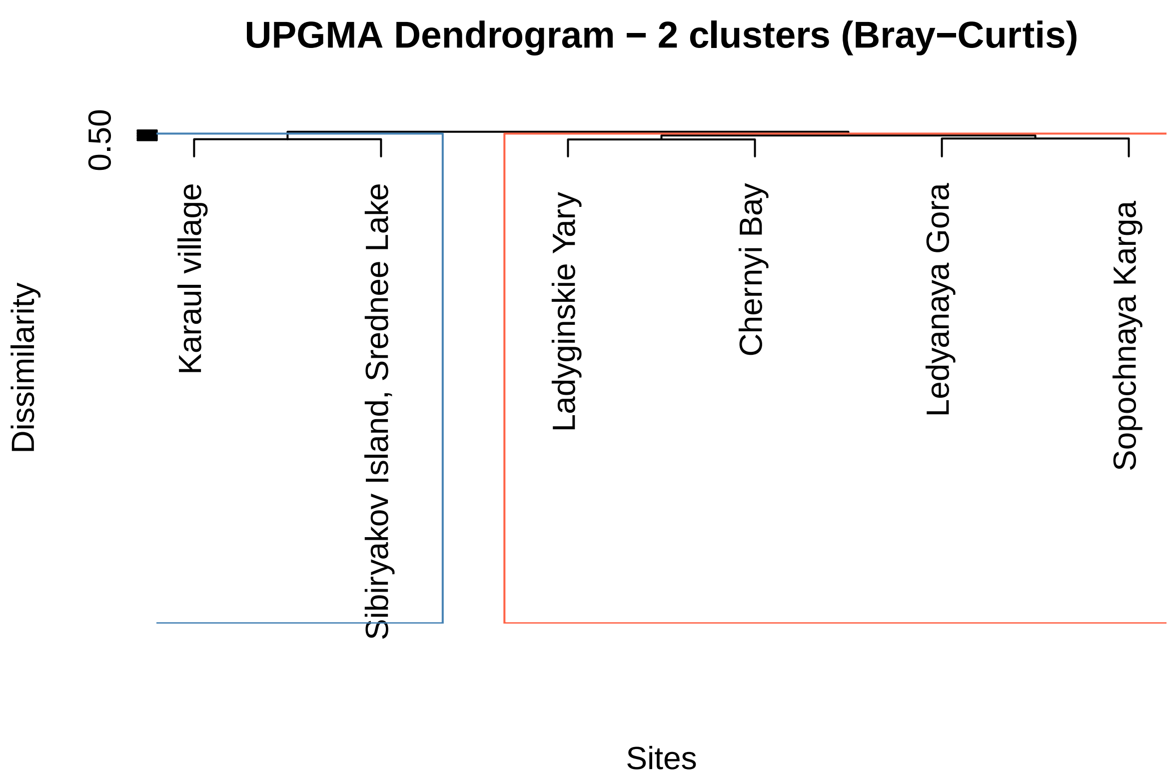
```r
k <- 2
clusters <- cutree(fit_clust, k = k)

cat("\nCluster membership (k =", k, "):\n")
```

```
##
## Cluster membership (k = 2 ):
```

```r
print(clusters)
```

```
##                  Ledyanaya Gora                  Karaul village
##                               1                               2
##                Ladyginskie Yary                Sopochnaya Karga
##                               1                               1
## Sibiryakov Island, Srednee Lake                     Chernyi Bay
##                               2                               1
```

```r
# Highlight clusters on dendrogram
plot(fit_clust, hang = -1,
     main = paste("UPGMA Dendrogram -", k, "clusters (Bray-Curtis)"),
     xlab = "Sites", sub = "", ylab = "Dissimilarity")
rect.hclust(fit_clust, k = k, border = c("steelblue", "tomato")[1:k])
```

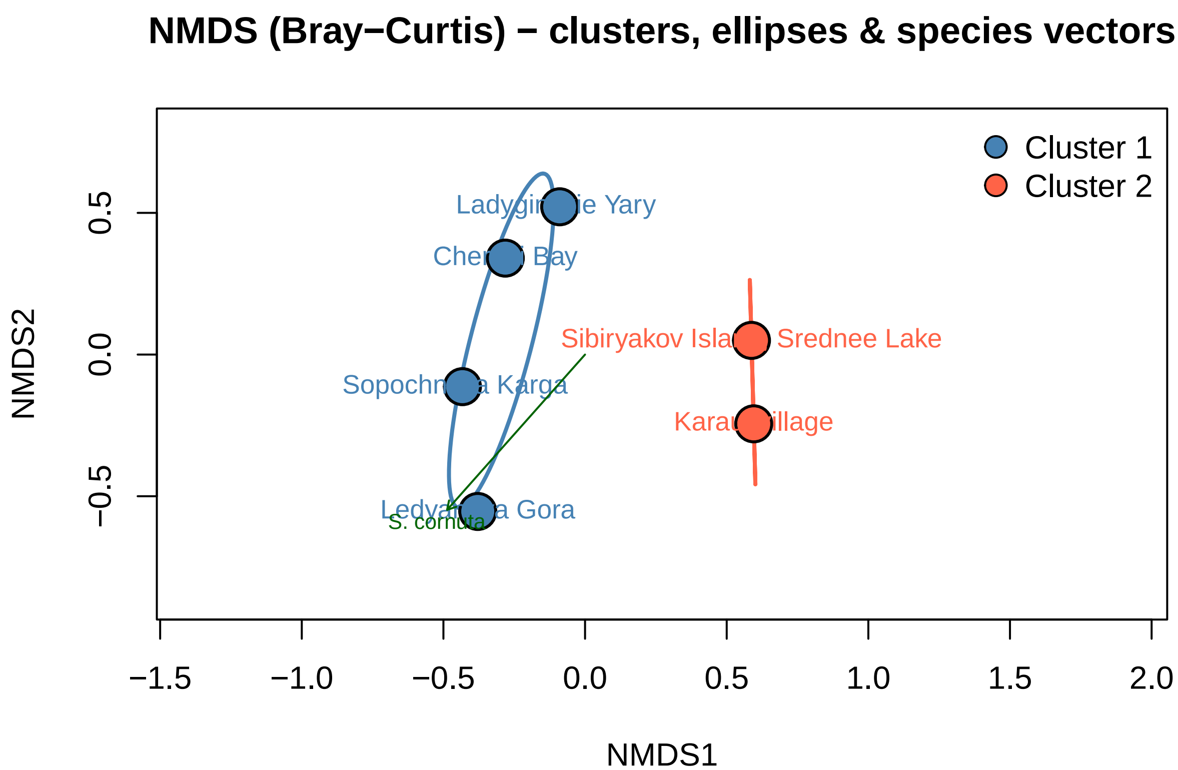



```
# STEP 5: NMDS plot with colours, ellipses, arrows, labels
```

```
cluster_cols <- c("steelblue", "tomato", "forestgreen")
site_cols    <- cluster_cols[clusters]

plot(ord, type = "n",
     main = "NMDS (Bray-Curtis) - clusters, ellipses & species vectors")

# 95% confidence ellipses per cluster
ordiellipse(ord, groups = clusters, kind = "se", conf = 0.95,
            col = cluster_cols, lwd = 2, label = FALSE)
```

```
## Warning in chol.default(cov, pivot = TRUE): the matrix is either rank-deficient
## or not positive definite
```

```
# Site points coloured by cluster
points(ord, display = "sites",
       pch = 21, bg = site_cols, col = "black", cex = 2.5, lwd = 1.5)

# Non-overlapping site labels
orditorp(ord, display = "sites",
         col = site_cols, cex = 0.8, air = 0.9, pch = NA)

# Significant species arrows (p <= 0.05)
plot(fit_sp, p.max = 0.05, col = "darkgreen", cex = 0.7, arrow.mul = 0.8, add = TRUE)

# Legend
legend("topright",
       legend = paste("Cluster", 1:k),
       pt.bg  = cluster_cols[1:k],
       pch = 21, pt.cex = 1.5, bty = "n")
```




```
# STEP 6: PERMANOVA - test whether clusters differ significantly
```

```
perm_result <- adonis2(d_bray ~ clusters, permutations = 999)
```

```
## 'nperm' >= set of all permutations: complete enumeration.
## Set of permutations < 'minperm'. Generating entire set.
```

```
cat("\n--- PERMANOVA results ---\n")
```

```
##
## --- PERMANOVA results ---
```

```
print(perm_result)
```

```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 719
##
## adonis2(formula = d_bray ~ clusters, permutations = 999)
##          Df SumOfSqs      R2      F  Pr(>F)
## Model     1  0.49182 0.40881 2.7661 0.06667 .
## Residual  4  0.71121 0.59119
## Total     5  1.20303 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
R2   <- perm_result$R2[1]
pval <- perm_result$`Pr(>F)`[1]

cat(sprintf("\nConclusion: PERMANOVA R2 = %.3f, p = %.3f\n", R2, pval))
```

```
##
```

```
## Conclusion: PERMANOVA R2 = 0.409, p = 0.067
```

```r
if (pval <= 0.05) {
  cat("The clusters differ SIGNIFICANTLY in species composition (p <= 0.05).\n")
} else {
  cat("No significant difference between clusters (p > 0.05).\n")
  cat("This is expected with only 6 sites - insufficient power for PERMANOVA.\n")
}
```

```
## No significant difference between clusters (p > 0.05).
## This is expected with only 6 sites - insufficient power for PERMANOVA.
```